MIL-G-51203A(MU)
17 October 1966
SUPERSEDING
MIL-G-51203(MU)
15 May 1964

MILITARY SPECIFICATION

GUN, FLAME THROWER, PORTABLE, M7

1. SCOPE

1.1 This specification covers one type of portable flame thrower gun.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-117 - Bags, Interior Packaging.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.

FSC 1040

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-810 - Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

US ARMY EDGEWOOD ARSENAL

DL81-1-4920 - Gun, Flame Thrower, Portable, M7.

(All drawings referenced in this specification are listed on Drawing DL81-1-4920.)

PUBLICATIONS

US ARMY EDGEWOOD ARSENAL

136-300-160 - Instruction Manual for the Installation, Operation and Maintenance of Tester, Ignition Head, Portable Flame Thrower Gun, Q56.
136-300-163 - Instruction Manual for the Installation, Operation and Maintenance of Tester, Hydrostatic, Flame Thrower Gun, Q59.
136-300-166 - Instruction Manual for the Operation of Test Device, Trigger Mechanism, Flame Thrower Gun, Q58.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL81-1-4920 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the gun shall conform to the specifications and drawings listed on Drawing DL81-1-4920 and subsidiary drawings.

3.2 Assembly. The gun shall be assembled as specified on Drawing D81-1-4920.

3.3 Leakage. The assembled gun shall not leak when tested in accordance with 4.3.4.1.

3.4 Safety.

3.4.1 Valve lever. The safety mechanism shall not release the valve lever when a round metal weight of 5 to 6 pounds is dropped on the valve lever from a minimum height of 1 foot, when tested in accordance with 4.3.4.2(a).

3.4.2 Ignition lever. The safety mechanism shall not release the ignition lever when a 50- to 51-pound dead weight is placed at the index finger position when tested in accordance with 4.3.4.2(b).

3.5 Functioning.

3.5.1 Valve lever. After release of the safety mechanism, the valve lever shall be depressed, when a 30- to 31-pound load is applied, so that the needle valve is retracted to a minimum of 3/8 inch from the end of the nozzle orifice and the needle valve shall not move when a 13- to 14-pound load is applied, when tested in accordance with 4.3.4.3(a).

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3.5.2 Ignition lever. After release of the safety mechanism, the ignition lever shall be depressed, when a 17 to 18-pound load is applied, so that the ignition pin extends a minimum of 1/4 inch from the face of the ignition head body and the ignition lever shall not move when a 3- to 3-1/2-pound load is applied when tested as specified in 4.3.4.3(b).

3.5.3 Nozzle shield. The nozzle shield shall freely screw onto and off of the ignition grip when tested as specified in 4.3.4.3(c). As the shield is screwed on, the latch shall engage in the ratchet notches in the nozzle shield and prevent return motion. The latch shall finally engage in the locking groove in the shield when the shield is screwed on hand tight.

3.5.4 Ignition head. The assembled ignition head shall extend the cylinder plunger of the simulated ignition cylinder (Q56 tester) a minimum of 13/64 of an inch and index the simulated ignition cylinder to the next cylinder plunger through five positions when tested as specified in 4.3.4.3(d). In addition, the ignition pin shall extend a minimum of 1/4 inch beyond the flat surface of the ignition head body, when ignition lever is fully depressed when tested as specified in 4.3.4.3(d).

3.5.5 Moving parts. The valve lever, valve safety lever, ignition lever, ignition safety lever, valve rod and ignition pin shall be free to move without sticking or binding.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of guns shall be produced in accordance with this specification for examination and test (see 4.2).

3.7 High temperature. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.1.

3.8 Low temperature. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.2.

3.9 Temperature shock. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.3.

3.10 Rain. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.4.

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3.11 Humidity (cycling). The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.5.

3.12 Salt fog. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.6.

3.13 Sand and dust. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.7.

3.14 Vibration. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.8.

3.15 Mechanical shock. The gun shall conform to 3.3, 3.4, and 3.5 when tested in accordance with 4.2.3.9.

3.16 Workmanship. The gun shall be free from foreign matter and damage such as dents, cracks, or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of 59 guns shall be produced using the same methods, material, and equipment that will be used in regular production.

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4.2.2 Inspection. The preproduction sample shall be tested in accordance with table I.

Table I. Preproduction sampling and inspection acceptance/rejection criteria

Quantity	Requirement	Test	Accept	Reject
10	3.7	4.2.3.1	0	1
10	3.8	4.2.3.2	0	1
10	3.9	4.2.3.3	0	1
10	3.10	4.2.3.4	0	1
3	3.11	4.2.3.5	0	1
3	3.12	4.2.3.6	0	1
3	3.13	4.2.3.7	0	1
5	3.14	4.2.3.8	0	1
5	3.15	4.2.3.9	0	1

4.2.3 Tests.

4.2.3.1 High temperature. The gun shall be tested in accordance with MIL-STD-810, method 501.1.

4.2.3.2 Low temperature. The gun shall be tested in accordance with MIL-STD-810, method 502.2, except that -40°F shall be substituted for -80°F.

4.2.3.3 Temperature shock. The gun shall be tested in accordance with MIL-STD-810, method 503.1.

4.2.3.4 Rain. The gun shall be tested in accordance with MIL-STD-810, method 506.1.

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4.2.3.5 Humidity (cycling). The gun shall be tested in accordance with MIL-STD-810, method 507.1.

4.2.3.6 Salt fog. The gun shall be tested in accordance with MIL-STD-810, method 509.1.

4.2.3.7 Sand and dust. The gun shall be tested in accordance with MIL-STD-810, method 510.1.

4.2.3.8 Vibration. The gun shall be securely fastened to the vibration table and vibrated in accordance with figure 514-6, curve A and time table 514-II, schedule IV of MIL-STD-810, method 514.1.

4.2.3.9 Mechanical shock. The gun shall be mounted on the test plate and subjected to three shocks in each direction applied along three mutually perpendicular axes for a total of 18 shocks. The shock pulse shape shall be in accordance with figure 516-1, with a peak value (A) of 15 g and a nominal duration (D) of 11 milliseconds in accordance with MIL-STD-810, method 516.1.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the guns produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.3.2 Sampling for examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection provisions.

4.3.3.1 For examination and nondestructive tests. Sample guns and the level A preparation for delivery thereof shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For critical defects. Each gun assembly in the lot shall be inspected for critical characteristics listed in the classification of defects.

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4.3.3.3 Classification of defects.

(a) Gun, flame thrower, portable, M7 (Dwg D81-1-4920).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.3.4.1
2	Safety	4.3.4.2
3	Functioning	4.3.4.3
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Ignition lever in normal position, ignition pin protrudes from ignition head body	
103	Ignition lever fully depressed, ignition pin extension from face of ignition head body incorrect	
104	Length of needle valve retraction incorrect	
105	Thread 3/4-14 NPT incorrect (Dwg C81-1-4948)	
106	Moving parts stick and bind	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing or incorrect	
202	Mating surfaces of ignition grip cover and ignition grip out of alignment (Dwg E81-1-4923 and E81-1-4924)	
203	Mating surfaces of valve grip cover and valve grip out of alignment (Dwg D81-1-4965 and D81-1-4966)	
204	Workmanship (see 3.16)	

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(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Ruptured bag
102	Loose pack
103	Strapping loose or missing
104	Marking illegible or incorrect

4.3.4 Tests.

4.3.4.1 Leakage. The gun shall be filled with preservative oil conforming to grade 2 or 3 of MIL-L-21260 and subjected to a pressure of 740 to 750 pounds per square inch gage (psig) for 60 to 70 seconds. The area around the needle valve, the gasket between the barrel assembly and the inlet body, the packing and the inlet body and hose inlet shall be observed for leakage. If no leaks are observed, the gun shall be placed in the Q59 tester as per instruction manual 136-300-163 and the pressure reduced to 340 to 350 psig. Three unignited bursts of a minimum 2-second duration shall be fired at an interval of 30 seconds maximum between bursts. Immediately after the final burst, the nozzle shall be wiped clean and the gun observed for leakage while pressurized for 60 to 70 seconds. The nozzle end shall be placed over a blotter or other surface paper and the assembly struck with the hand. If moisture is deposited on the blotter or paper, the assembly shall be considered a leaker. A test dye may be used in the test fuel to improve contrast and facilitate leak detection.

4.3.4.2 Safety.

(a) Valve lever. The valve lever shall be tested using the Q58 tester as per instruction manual 136-300-166. The assembled gun shall be clamped in a vertical position with the nozzle up and with the safety valve lever not depressed. The round metal weight shall strike only the valve lever at the index finger position at approximately 90 degrees to the grip surface and in the direction of the lever travel.

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(b) Ignition lever. The ignition lever shall be tested using the Q58 tester as per instruction manual 136-300-166. The assembled gun shall be clamped in a horizontal position with the ignition lever up and the safety lever not depressed. The weight placed at the index finger position shall be applied at approximately 90 degrees to the grip surface and in the direction of lever travel. A hook may be used to apply load.

4.3.4.3 Functioning.

(a) Valve lever. The valve lever shall be tested using the Q58 tester as per instruction manual 136-300-166. The assembled gun shall be fixed in a vertical position with the nozzle up. The safety shall be released and the specified load slowly applied to the index finger notch. The load shall be removed and the valve lever retested with the 13- to 14-pound load.

(b) Ignition lever. The ignition lever shall be tested using the Q58 tester as per instruction manual 136-300-166. The assembled gun less the ignition cylinder shall be placed in a horizontal position with the ignition lever up and the specified load applied. The safety shall be released and the lever retested with the 3- to 3-1/2-pound load.

(c) Nozzle shield. The nozzle shield shall be screwed onto the ignition grip and the latch engaged in the ratchet. The latch shall engage the locking groove when the shield is hand tight and prevent the nozzle shield from returning. The latch shall be disengaged from the locking groove and the nozzle shield removed.

(d) Ignition head. The ignition head shall be tested using the Q56 tester per instruction manual 136-300-160, except that the amount of travel for the cylinder plunger (6c) (7) shall conform to 3.5.4 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. Unpainted metal surfaces including threaded areas of the gun shall be cleaned in accordance with MIL-P-116, using cleaning method C-3 and any of the drying methods listed therein.

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5.2 Preservation and packaging.

5.2.1 Level A. The gun, cleaned and dried as specified in 5.1, shall have all unpainted surfaces including threaded areas coated with preservative oil conforming to grade 2 or 3 of MIL-L-21260. All excess oil shall be drained or wiped off. The gun shall then be placed in a type I, class e heat sealable bag conforming to MIL-B-117. Before the bag is sealed, sufficient air shall be withdrawn so that the bag conforms to the shape of the gun but is not damaged or ruptured. Each gun shall be individually packaged in accordance with figure 1 in a W6c, style RSC fiberboard box conforming to PPP-B-636, having approximate inside dimensions of 7-5/8 by 4-1/4 by 21-1/2 inches. Fiberboard cells in figure 1 shall be fabricated from the same material as the box. Filler pads of material of the same type as the box shall be used as necessary to restrict free movement of the gun. The box shall be closed and all seams and joints shall be sealed with tape in accordance with the appendix to PPP-B-636.

5.2.2 Level C. Packaging shall be in accordance with 5.2.1 except the barrier bag is not required and the box and cell material may be of domestic grade material. Sealing of the seams and joints is not required.

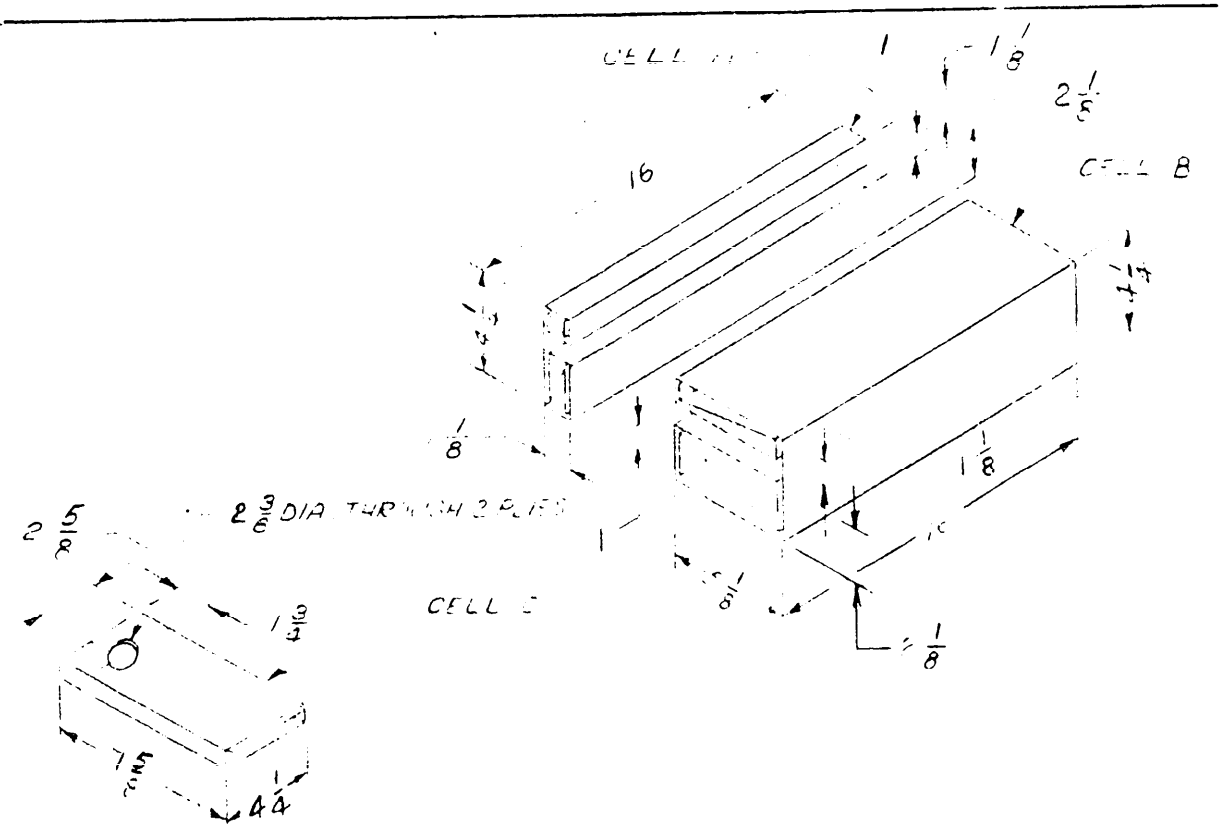
5.3 Packing.

5.3.1 Level A. Twenty-five guns, packaged as specified in 5.2.1, shall be packed together in five layers, five to a layer, in a snug-fitting class 2, style 2 or 2-1/2 wood box conforming to PPP-B-621. The box shall be closed in accordance with the requirements of the box specification, with strapping to be in accordance with the appendix thereto, using galvanized strapping.

5.3.2 Level B. Twenty-five guns, packaged as specified in 5.2.1, shall be packed as specified in 5.3.1 except that domestic type containers and coated-finish strapping shall be used.

5.3.3 Level C. A uniform number of guns, packaged as specified in 5.2.2, shall be packed in a manner to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

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GUN, PORTABLE FLAME THROWER
M 7

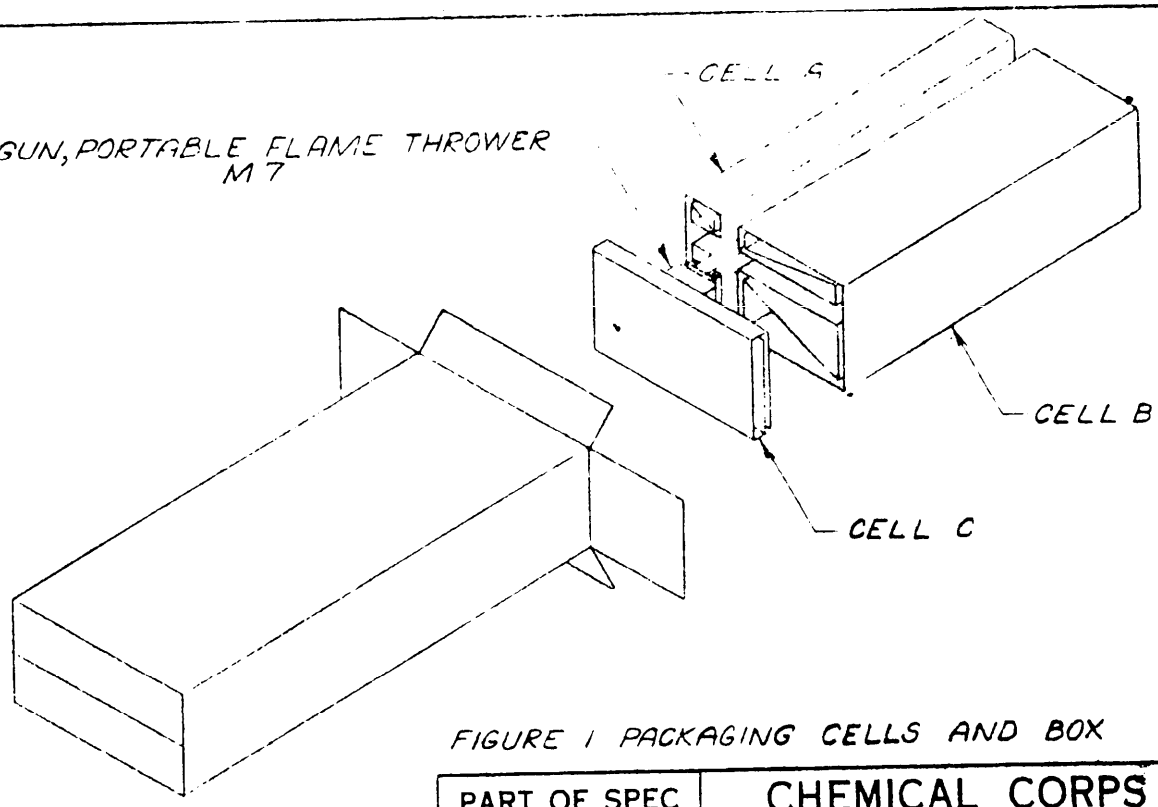


FIGURE 1 PACKAGING CELLS AND BOX

PART OF SPEC	CHEMICAL CORPS SPECIFICATION FIGURE
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5.4 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.5 Repair parts. When this item is being procured for stockage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair parts packaging data sheet (see 6.5).

6. NOTES

6.1 Intended use. This item is intended for use with a portable flame thrower. It controls firing and ignites the thickened fuel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Level of packaging and packing required.

(c) If packaging data sheet is required.

(d) If the Q56, Q58, and Q59 testers will be furnished by the Government.

(e) Preproduction.

(1) Time allowed for suppliers to submit samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Leakage. If leakage occurs at the valve seat, the valve rod may be rotated with a screwdriver. This rotation should generate a seat and effect a good seal.

6.4 Inlet hose adapter. The adapter required for the leakage test should be of a design which will perform the functions of the M8 fuel hose (Dwg C81-1-7001).

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6.5 Packaging data sheet. The packaging data sheet is identified by the Federal stock number for this item and should be ordered from the Commanding Officer, US Army Edgewood Arsenal, ATTN: SMUEA-TSE-DD, Edgewood Arsenal, Maryland 21010.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 1040-A059

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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OFFICIAL BUSINESS

COMMANDING OFFICER
EDGEWOOD ARSENAL
ATTN: SMUEA-QA-WS-S
EDGEWOOD ARSENAL, MD. 21010

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